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“Break Our Steganographic System” — the ins and outs of organizing BOSS

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18 may 2011



Outline

1 Introduction

2 Our Steganographic System

3 The outs



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Previous Data-hiding contests

Watermarking

- Break Our Watermarking System - 2006
- Break Our Watermarking System - 2nd Ed. - 2008
- Benchmark the robustness and security of Watermarking systems

Forensics etc...

- First IEEE Workitorial on Vision of the Unseen (WVU'08)
- Detection of forgeries, authentication, fingerprinting, data-hiding



From BOWS to BOSS

Motivations

- Benchmark content adaptive steganography
- Motivate new approaches in Steganalysis
- Benchmark steganalysis schemes in a fair and reproducible way
- Have fun!



Requirements and rules

Requirements and rules

- **Simple rules:** Ranking using detection accuracy, 1000 test images, half of them stego
- **Give time :** 2 months of warm-up, 4 months of competition
- **Give materials:** Source codes, state-of the art features extractors, 9074 training images
- **Prevent hacks:** One prediction per IP every three days to prevent Oracle attacks

BOSSBase - The training set

Camera model	# of images in <i>BOSSBase</i>
Leica M9	2267
Canon EOS DIGITAL REBEL XSi	1607
PENTAX K20D	1398
Canon EOS 400D DIGITAL	1354
Canon EOS 7D	1354
NIKON D70	1033
Canon EOS 40D	61

Corresponding Raw and EXIF files available too!

BOSSRank - The testing set

Camera model	# of images in BOSSRank
Leica M9	847
Panasonic Lumix DMC-FZ50	153

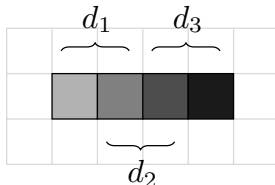
Random selection of Stego and Cover files.



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"Statistical" distortion

- Weights derived from differences of neighbouring pixels (d_1, d_2, d_3)

$$w(d_1, d_2, d_3) = \left(\sqrt{d_1^2 + d_2^2 + d_3^2} + \sigma \right)^{-\gamma} \quad (1)$$

defined for $d_i \in [-90; 90]$



Our Steganographic System (II/IV)

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agent
technology
center

Local distortion D_i computed using

- The weights $w(d_1, d_2, d_3)$ before and after altering one pixel
- The differences between averaged co-occurrence matrices for different directions

$$D(\mathbf{X}, \mathbf{Y}) = \sum_{d_1, d_2, d_3}^T -T \left[w(d_1, d_2, d_3) \left| F_{d_1, d_2, d_3}^{\mathbf{X}} - F_{d_1, d_2, d_3}^{\mathbf{Y}} \right| + w(d_1, d_2, d_3) \left| G_{d_1, d_2, d_3}^{\mathbf{X}} - G_{d_1, d_2, d_3}^{\mathbf{Y}} \right| \right],$$

Probability to modify one pixel i

$$p_i = \frac{1}{Z} e^{-\lambda D_i}. \quad (2)$$



Our Steganographic System (III/IV)

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Embedding

- Use Syndrome-Trellis Codes to minimise the distortion
- or simulate using p_i

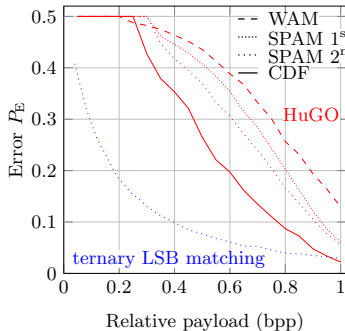
Last step

- Model correction: switch to $+1$ or -1 in order to minimise the global distortion



Our Steganographic System (IV/IV)

Benchmark with different feature sets:



Name: Highly Undetectable steGO (HUGO)



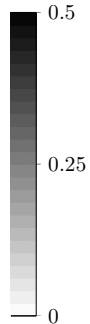
Where HUGO is Hidden?



(a) 0.25 bpp



(b) 0.5 bpp





Outline

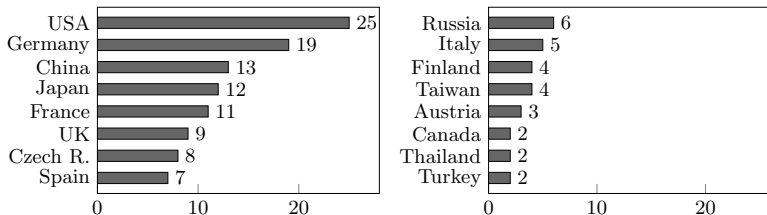
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Participation



of times BOSSRank has been downloaded per country



The final scores obtained using all the images of BOSSRank (1) are:

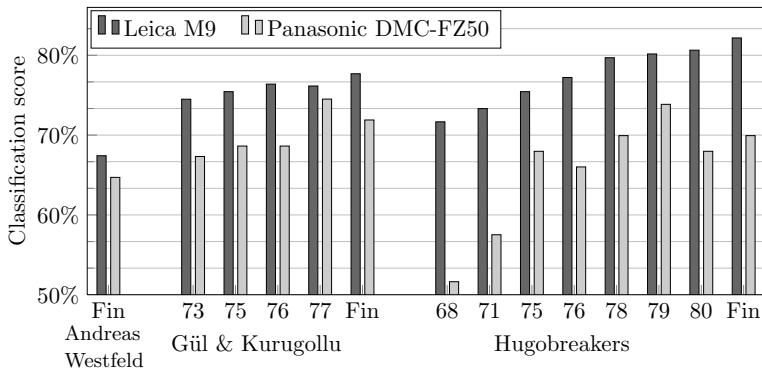
1) *HugoBreakers*: 80.3%

2) *Gül&Kurugollu*: 76.8%

3) *Andreas Westfeld*: 67.0%



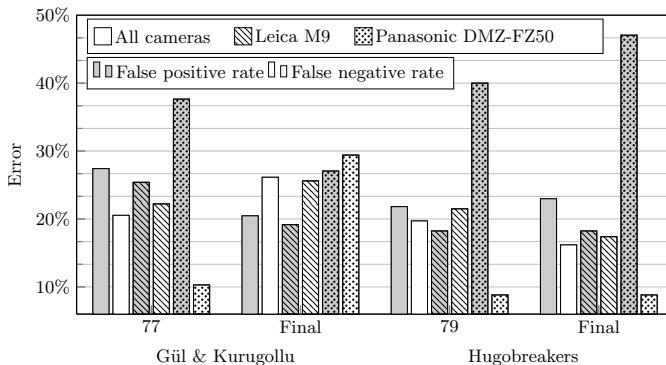
Results per camera



Images coming from the “unknown” camera are more difficult to classify: **cover-source mismatch problem.**



Results per type of images/camera

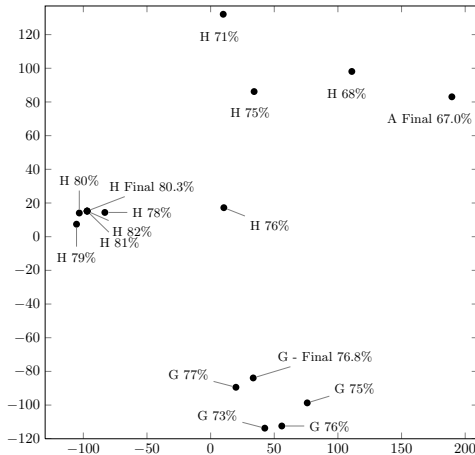


False positive and false negative rates according to the camera for the four best submissions.

Difficulties to classify cover images from the DMZ-FZ50.



Same strategies?



Multi-Dimensional Scaling plot



Conclusions and perspectives

Conclusions

- From 65% to 80.3%
- Raised important problems:
 - cover-source mismatch
 - designing new features
 - dealing with important number of features

Perspectives

- BOSSBase 1.0 will be available very soon (10000 images)
- Open BOSS as a benchmark tool for other steganographic schemes: LSB replacement, LSB matching, new stego schemes
- ...